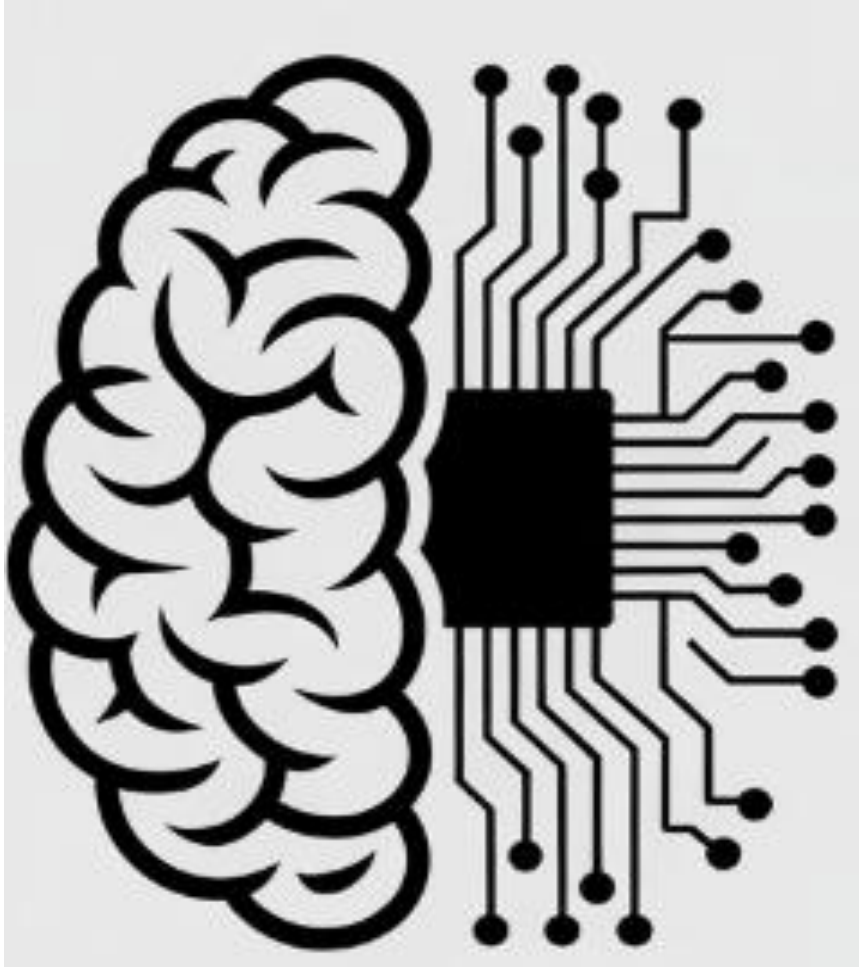




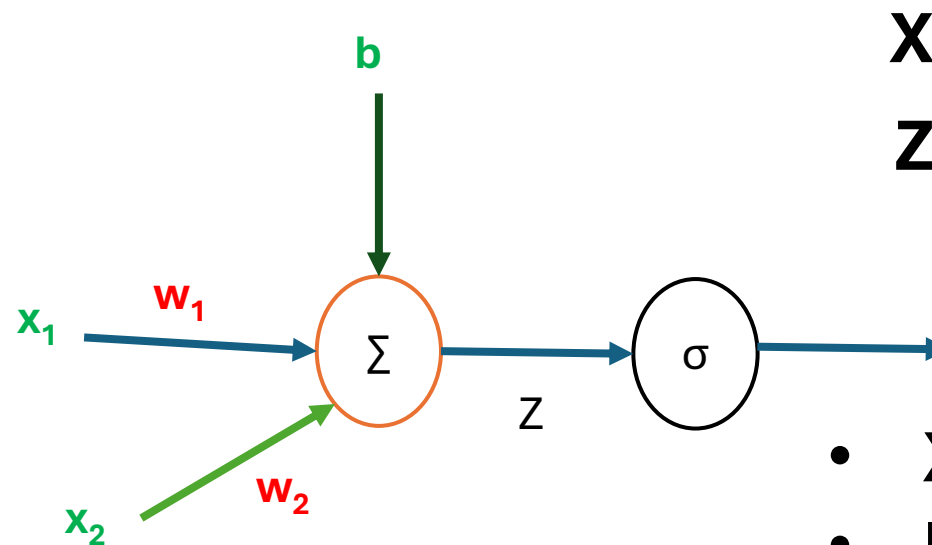
Neural Network

Dr. Sobia Tariq Javed

Suppose we have 2 inputs X_1 and X_2 with bias b .

Perceptron Learning

- **Perceptron Learning** is a supervised learning algorithm used for **binary classification**.
- It is used to classify data that is linearly separable.



$$X_0 = b$$

$$Z = X_0 + X_1W_1 + X_2W_2$$

- X_1 and X_2 are inputs
- b is called bias
- W_1 and W_2 are weights
- σ is the activation function

Suppose we have 2 inputs X_1 and X_2 with bias b .

Perceptron Learning

- $Z = X_1W_1 + X_2W_2 + b$

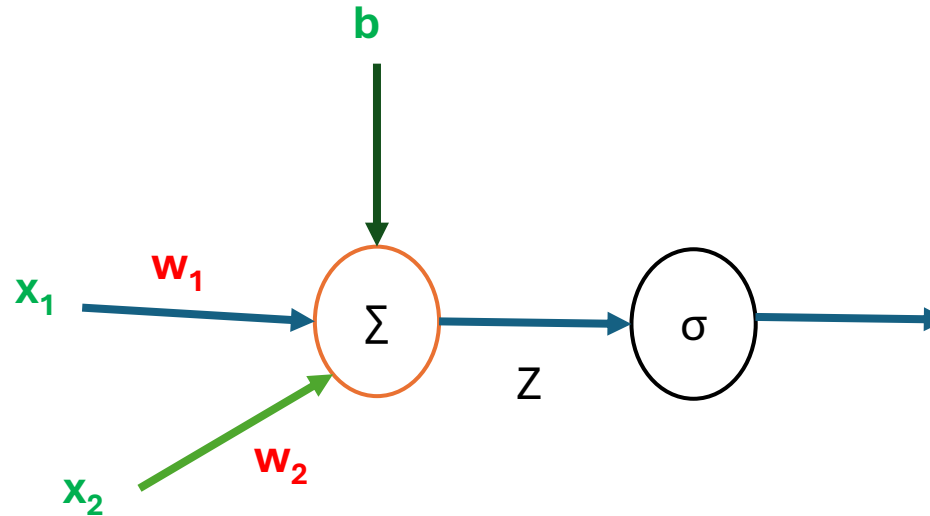
$$W = \begin{bmatrix} W_1 \\ W_2 \end{bmatrix}$$

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

- $Z = W^T X + b$

- $\sigma(Z) = \sigma(W^T X + b)$

- σ is the activation function



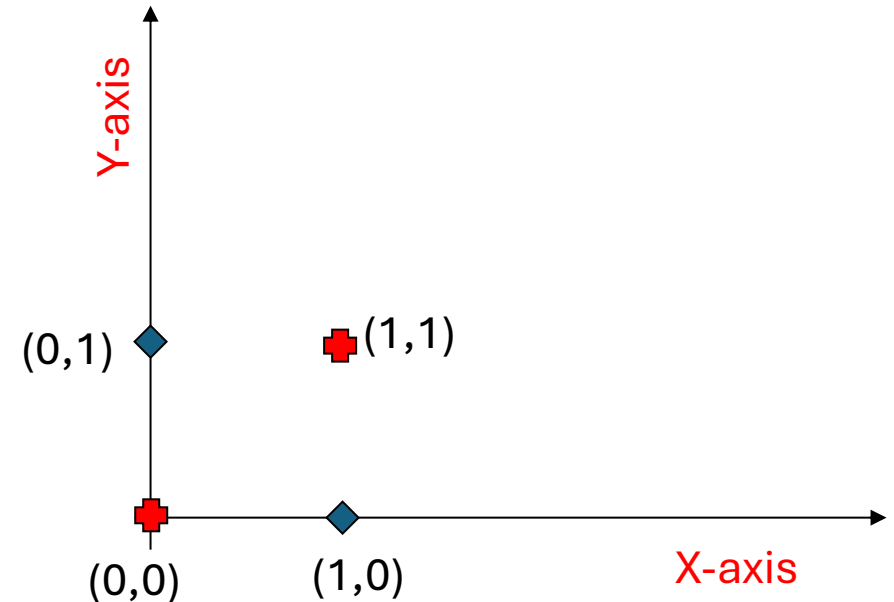
- X_1 and X_2 are inputs
- b is called bias
- W_1 and W_2 are weights

XOR Gate

$$Z = X_0W_0 + X_1W_1 + X_2W_2$$
$$W_0=0, W_1=W_2=1$$

- A typical XOR gate truth table is shown below

Input		Actual Output	Z
X	Y		
0	0	0	
0	1	1	
1	0	1	
1	1	0	

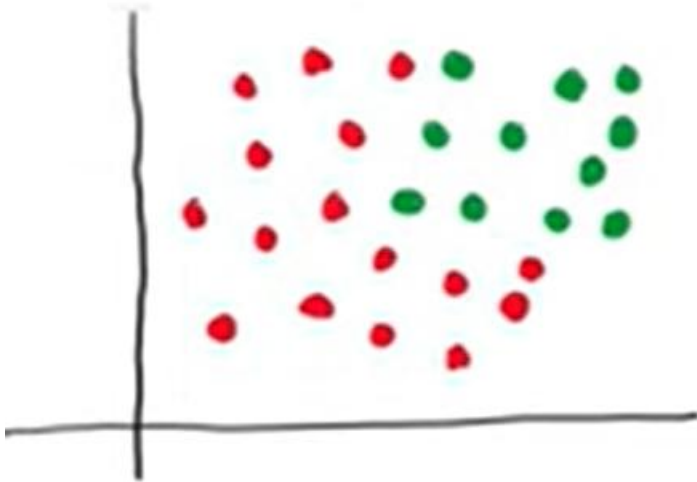


Multi-layer Perceptron

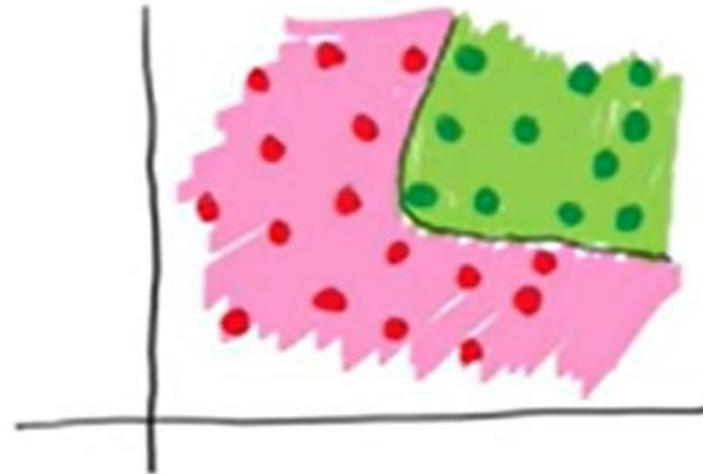
- How to separate the data using straight lines?

● Class A

● Class B



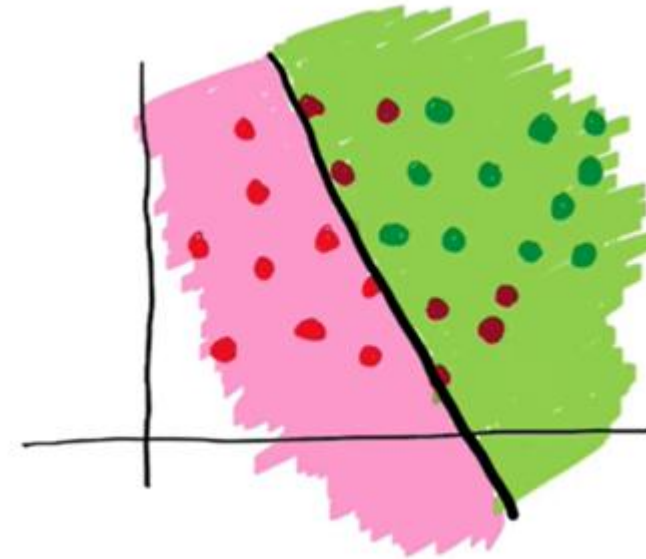
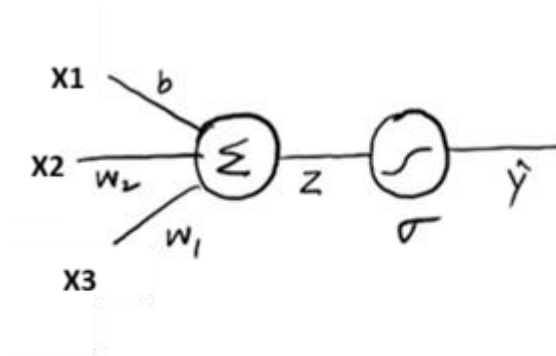
Curve required



Multi-layer Perceptron

- Class A
- Class B

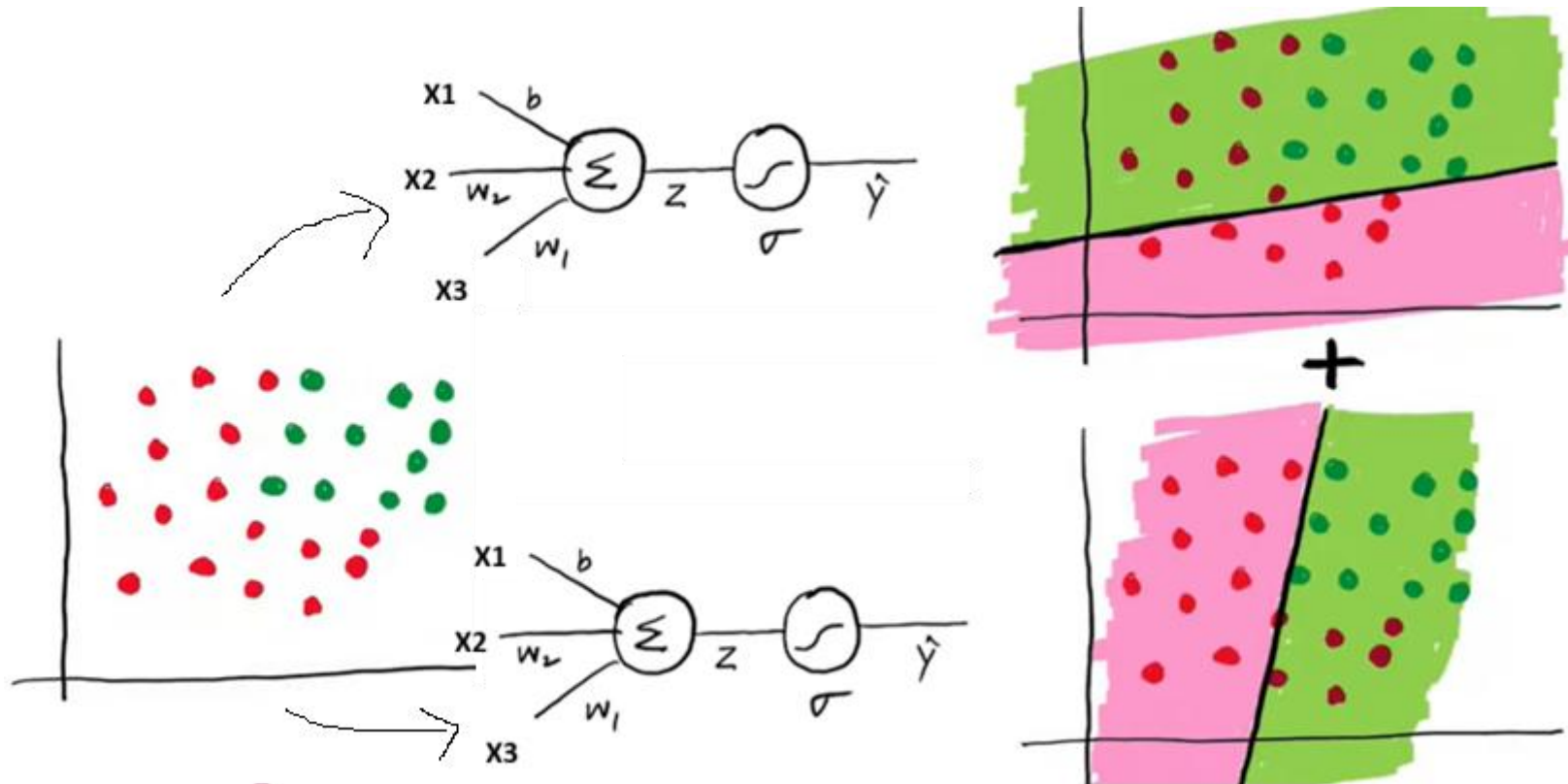
- With single perceptron



Multi-layer Perceptron

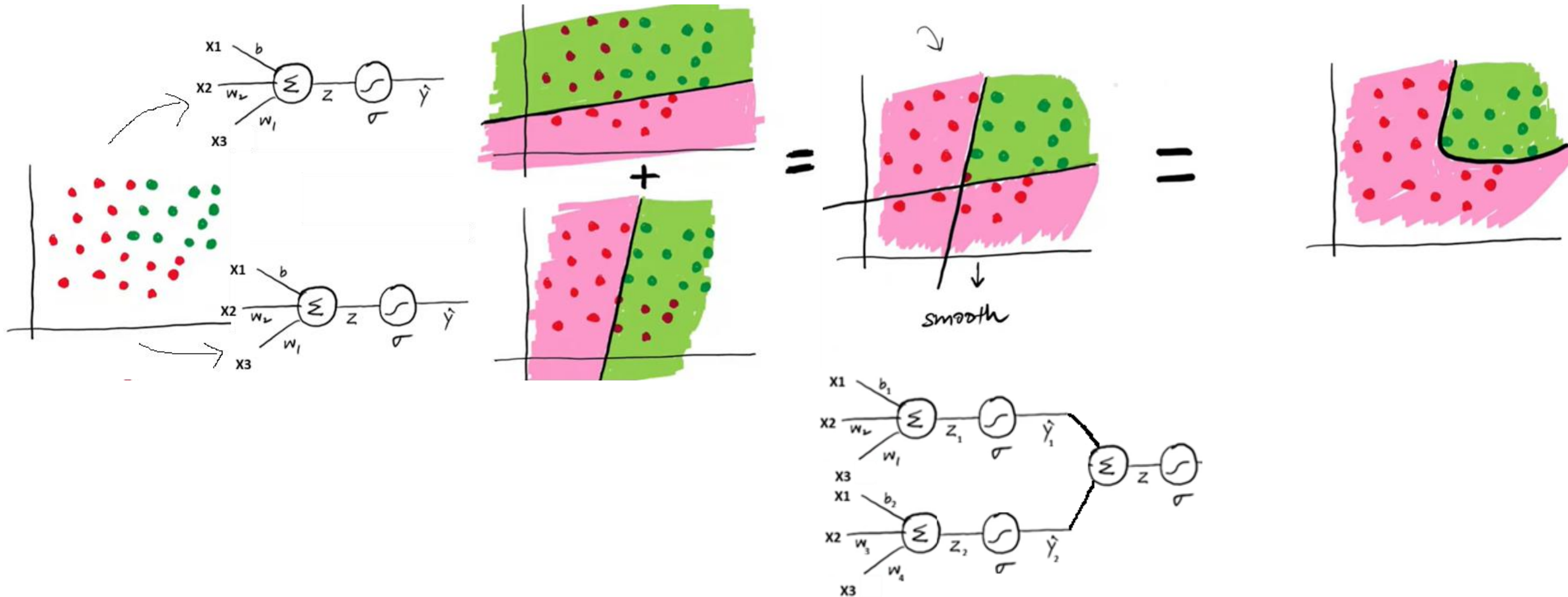
- Class A
- Class B

- With two perceptron



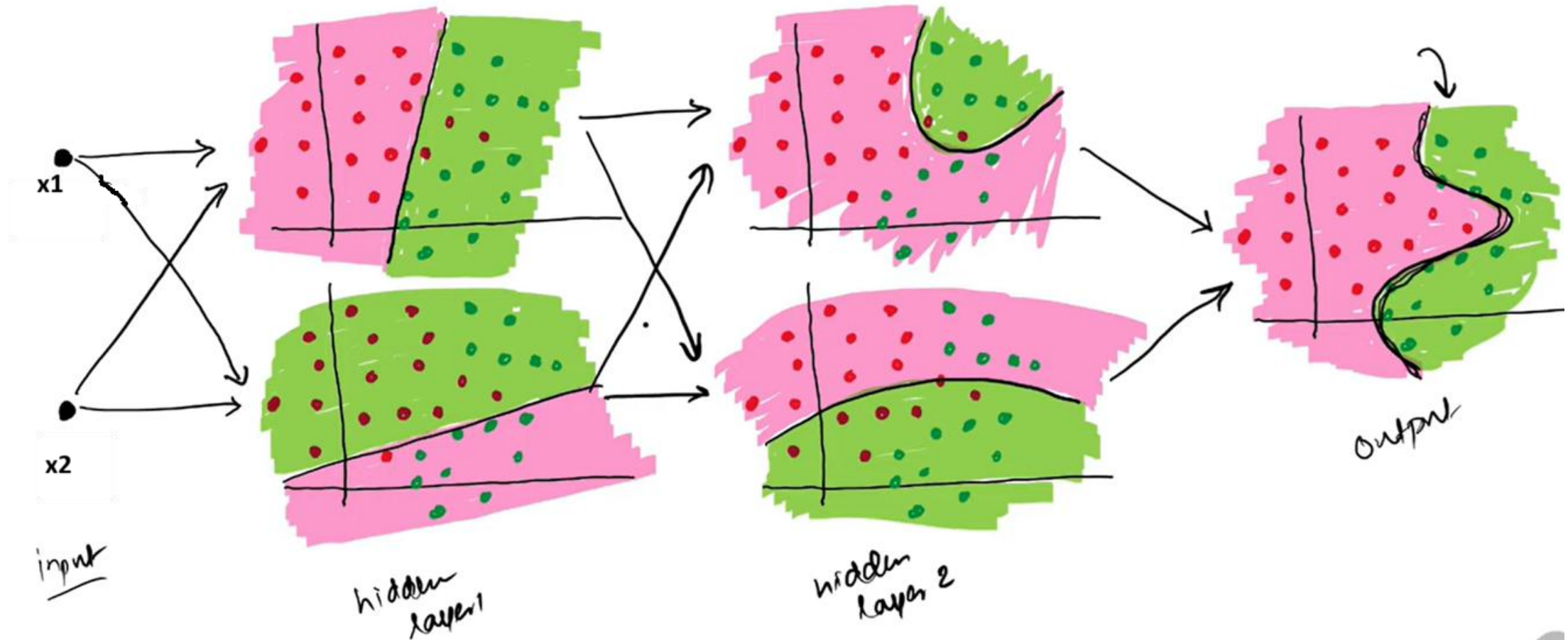
Multi-layer Perceptron

- Class A
- Class B



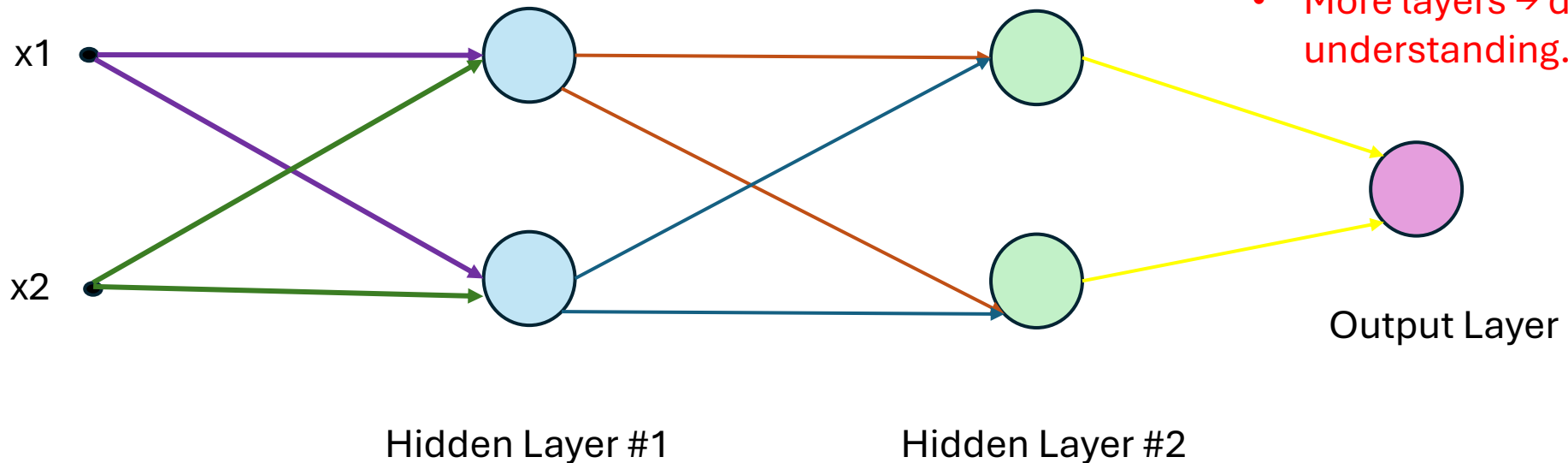
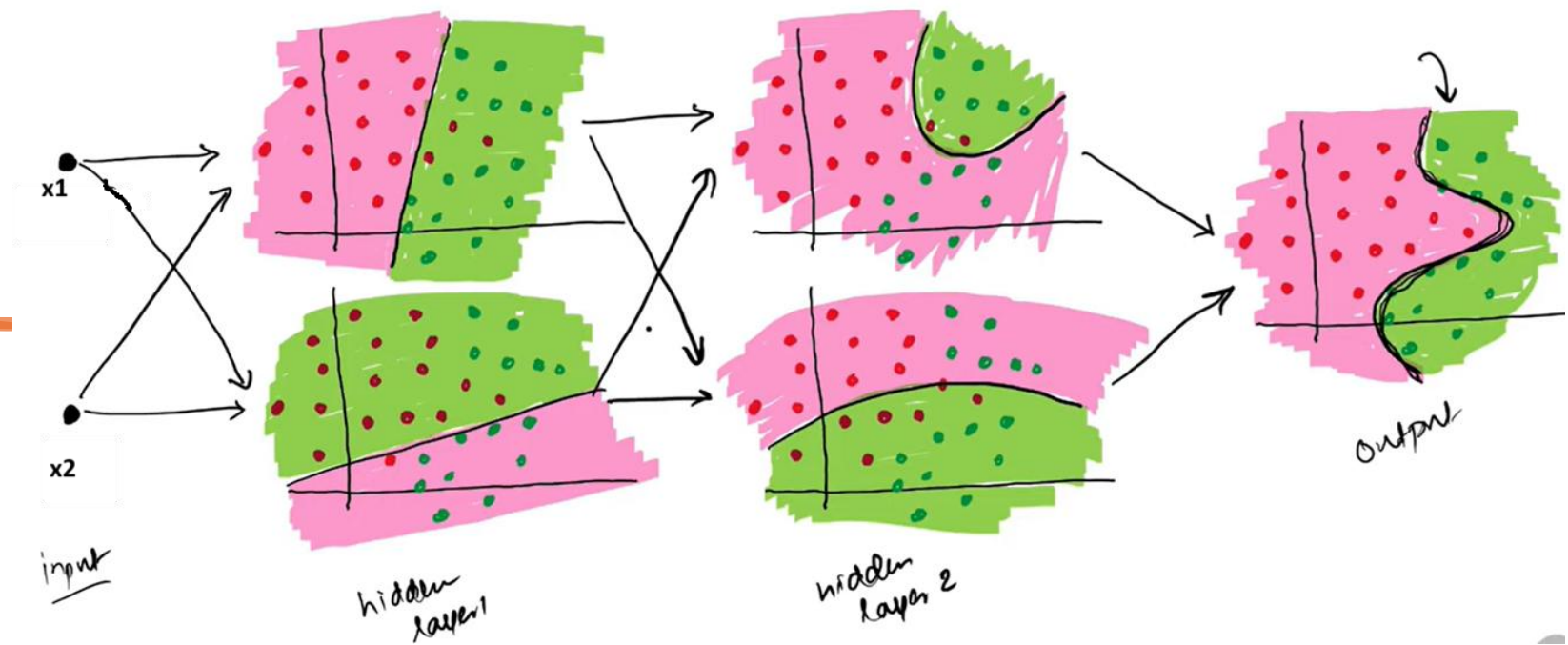
Multi-layer Perceptron

- Class A
- Class B



Multi-layer Perceptron

- Class A
- Class B



- Think of layers as **levels of understanding**.
- More layers $\rightarrow$ deeper understanding.

Adding Neurons Vs Adding Hidden Layer

- **Adding a neuron** in a hidden layer
 - increases the **width** of the network, allowing it to learn more features at the same level.
- **Adding a layer**
 - increases the depth of the network, enabling it to learn more complex and hierarchical patterns.

Adding Neurons Vs Adding Hidden Layer



- In short, neurons improve feature capacity, while layers improve abstraction and complexity of learning.

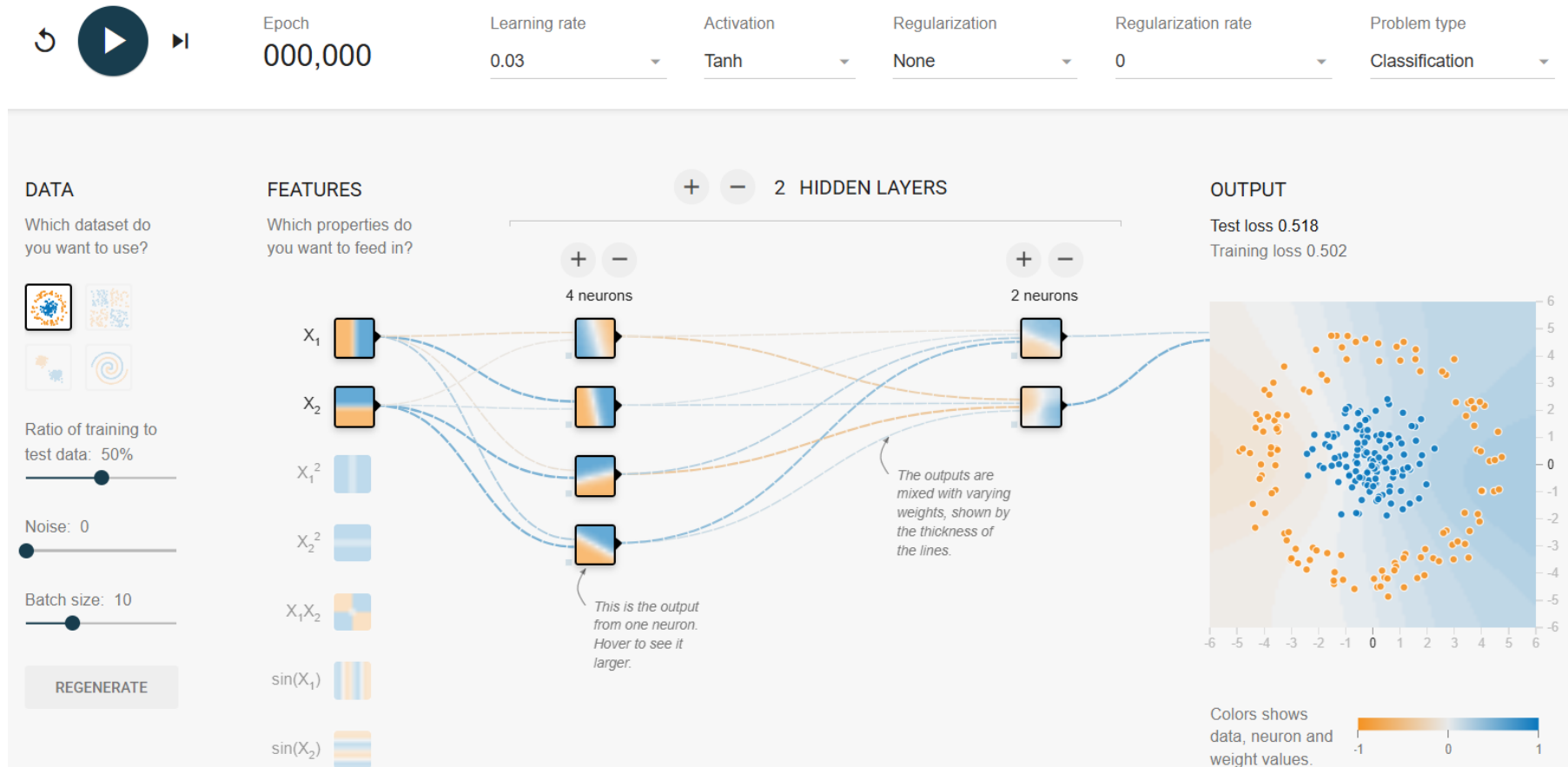
Example: Image Recognition

Layer	What it learns
Input layer	pixels
Hidden layer 1	edges
Hidden layer 2	shapes
Hidden layer 3	objects
Output	label

More layers → deeper understanding.

Try

- Tensorflow Playground - <https://playground.tensorflow.org/>



Neural Network

w_{ij}^k

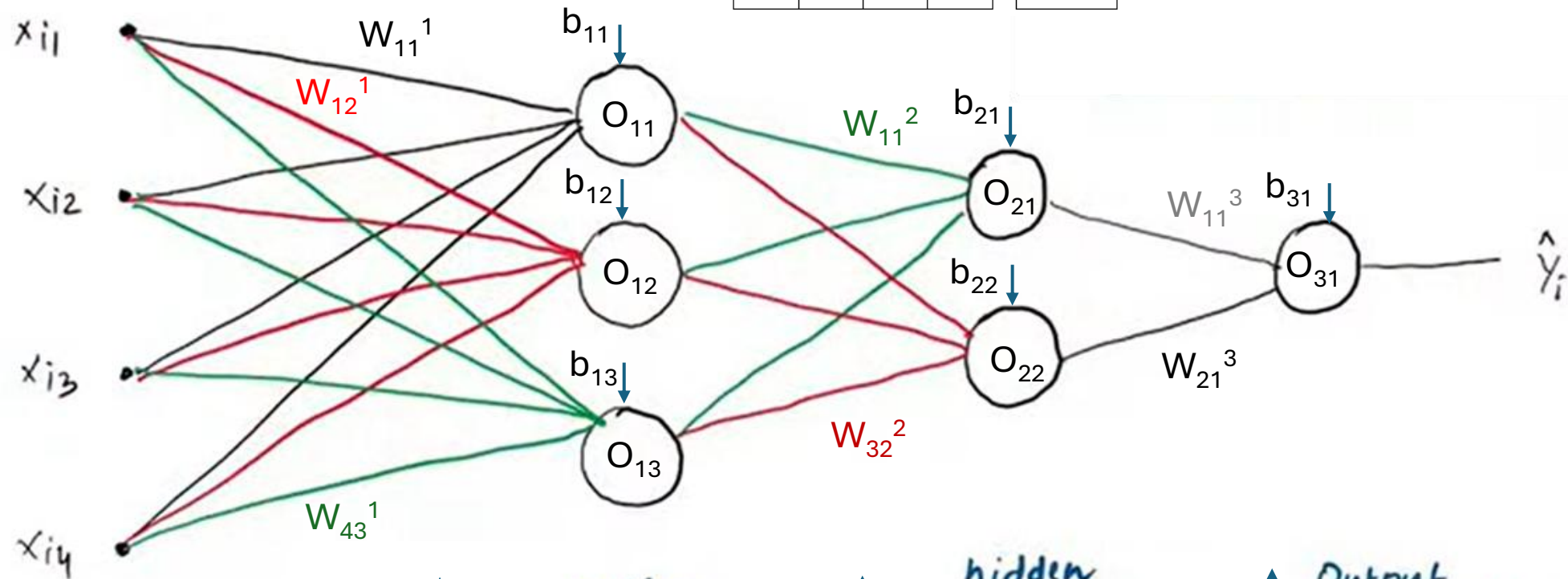
	A ₁	A ₂	A ₃	A ₄	Labels
X ₁	X ₁₁	X ₁₂	X ₁₃	X ₁₄	L ₁
X ₂	X ₂₁	X ₂₂	X ₂₃	X ₂₄	L ₂
X ₃	X ₃₁	X ₃₂	X ₃₃	X ₃₄	L ₃
⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮
X _m	X _{m1}	X _{m2}	X _{m3}	X _{m4}	L _m

Data {m×n}

m = # rows

n = 4

Each row is denoted by X_i



input layer (L0)

12 weights
3 bias

hidden layer #1 (L1)

6 weights
2 bias

hidden layer #2 (L2)

2 weights
1 bias

Output layer (L3)

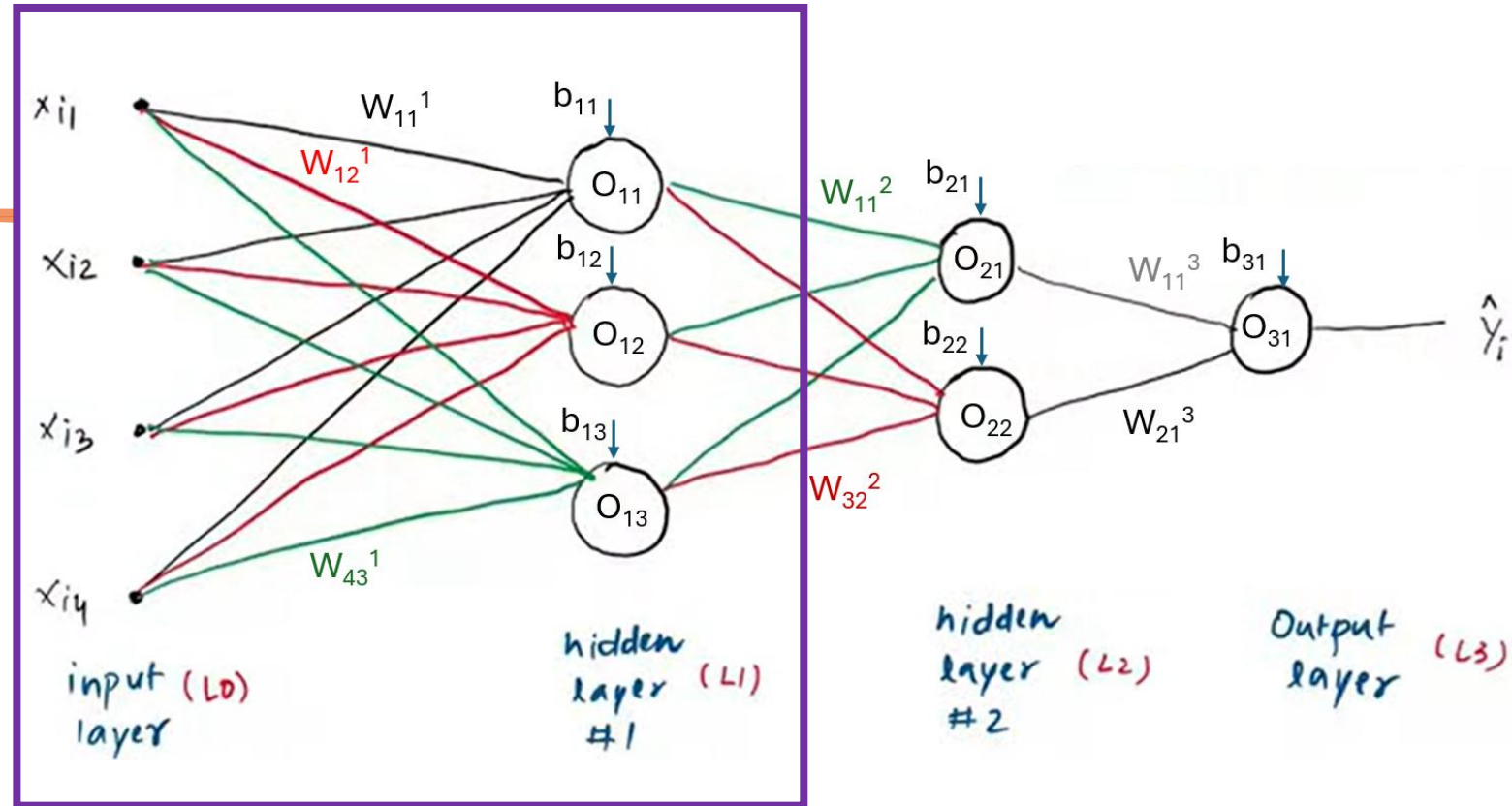
26 trainable parameters

Forward Pass- Layer 1

$$W^{[1]} = \begin{bmatrix} W_{11}^1 & W_{12}^1 & W_{13}^1 \\ W_{21}^1 & W_{22}^1 & W_{23}^1 \\ W_{31}^1 & W_{32}^1 & W_{33}^1 \\ W_{41}^1 & W_{42}^1 & W_{43}^1 \end{bmatrix} \quad X = a^{[0]} = \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} \quad b^{[1]} = \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$\begin{bmatrix} W_{11}^1 & W_{12}^1 & W_{13}^1 \\ W_{21}^1 & W_{22}^1 & W_{23}^1 \\ W_{31}^1 & W_{32}^1 & W_{33}^1 \\ W_{41}^1 & W_{42}^1 & W_{43}^1 \end{bmatrix}^T \cdot \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

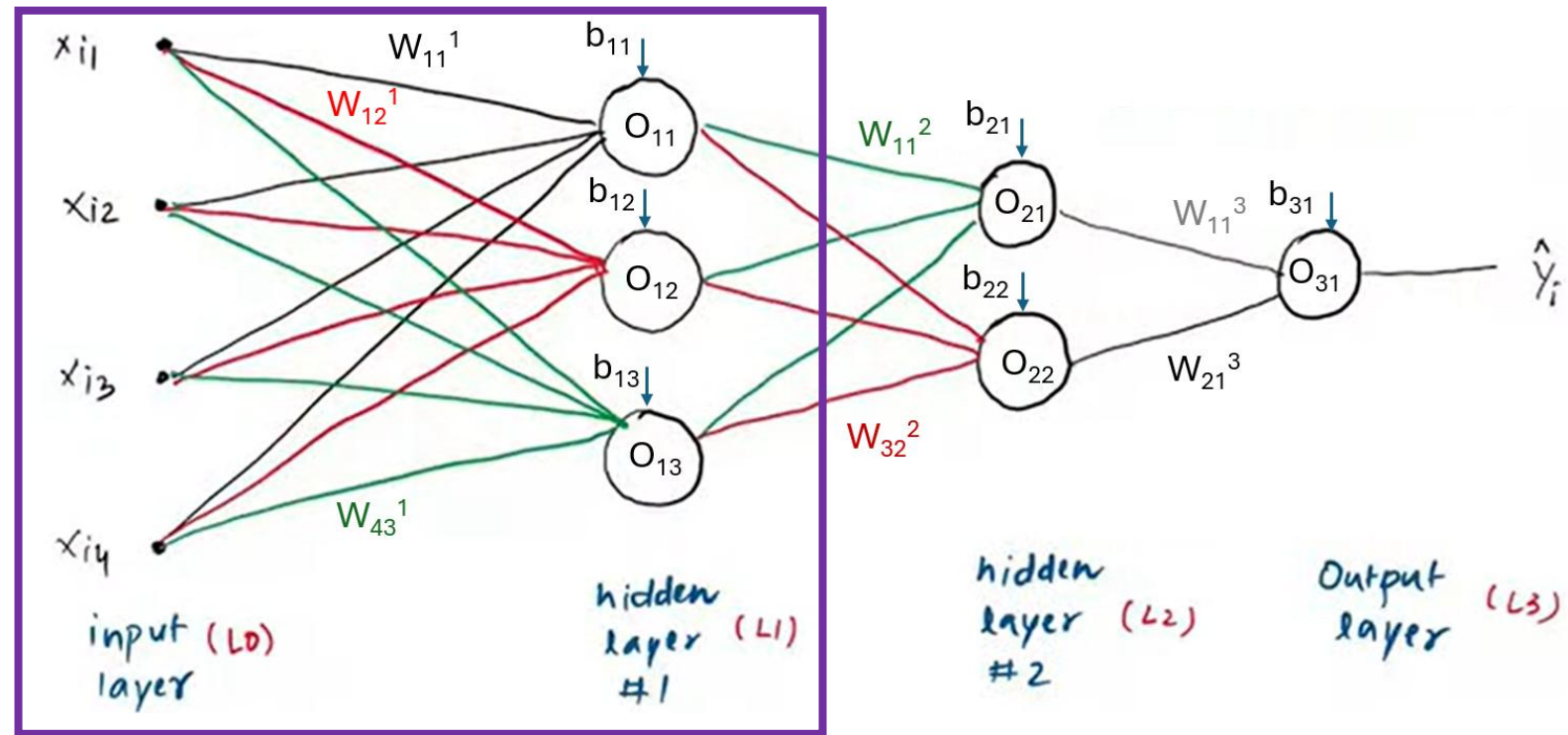
$$W^T X + b$$



$$\begin{bmatrix} W_{11}^1 & W_{12}^1 & W_{13}^1 \\ W_{21}^1 & W_{22}^1 & W_{23}^1 \\ W_{31}^1 & W_{32}^1 & W_{33}^1 \\ W_{41}^1 & W_{42}^1 & W_{43}^1 \end{bmatrix}^T \cdot \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

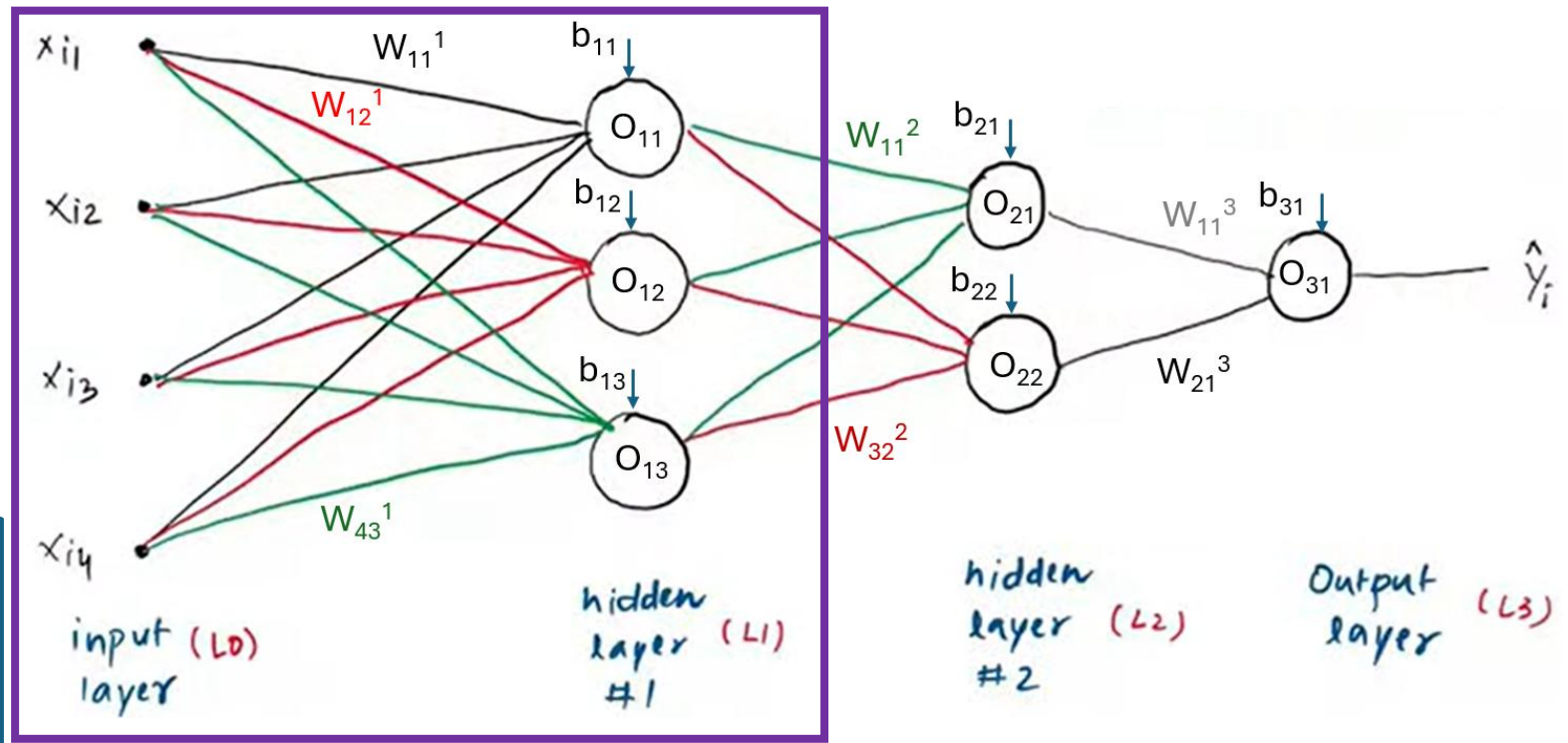
$$= \begin{bmatrix} W_{11}^1 & W_{21}^1 & W_{31}^1 & W_{41}^1 \\ W_{12}^1 & W_{22}^1 & W_{32}^1 & W_{42}^1 \\ W_{13}^1 & W_{23}^1 & W_{33}^1 & W_{43}^1 \end{bmatrix} \cdot \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$\underbrace{\quad (3 \times 4) \quad \cdot \quad (4 \times 1) \quad + \quad (3 \times 1) \quad}_{(3 \times 1) \leftarrow \text{Result}}$



$$= \begin{bmatrix} W_{11}^1 & W_{21}^1 & W_{31}^1 & W_{41}^1 \\ W_{12}^1 & W_{22}^1 & W_{32}^1 & W_{42}^1 \\ W_{13}^1 & W_{23}^1 & W_{33}^1 & W_{43}^1 \end{bmatrix} \cdot \begin{bmatrix} X_{i1} \\ X_{i2} \\ X_{i3} \\ X_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$= \begin{bmatrix} W_{11}^1 X_{i1} + W_{21}^1 X_{i2} + W_{31}^1 X_{i3} + W_{41}^1 X_{i4} \\ W_{12}^1 X_{i1} + W_{22}^1 X_{i2} + W_{32}^1 X_{i3} + W_{42}^1 X_{i4} \\ W_{13}^1 X_{i1} + W_{23}^1 X_{i2} + W_{33}^1 X_{i3} + W_{43}^1 X_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$



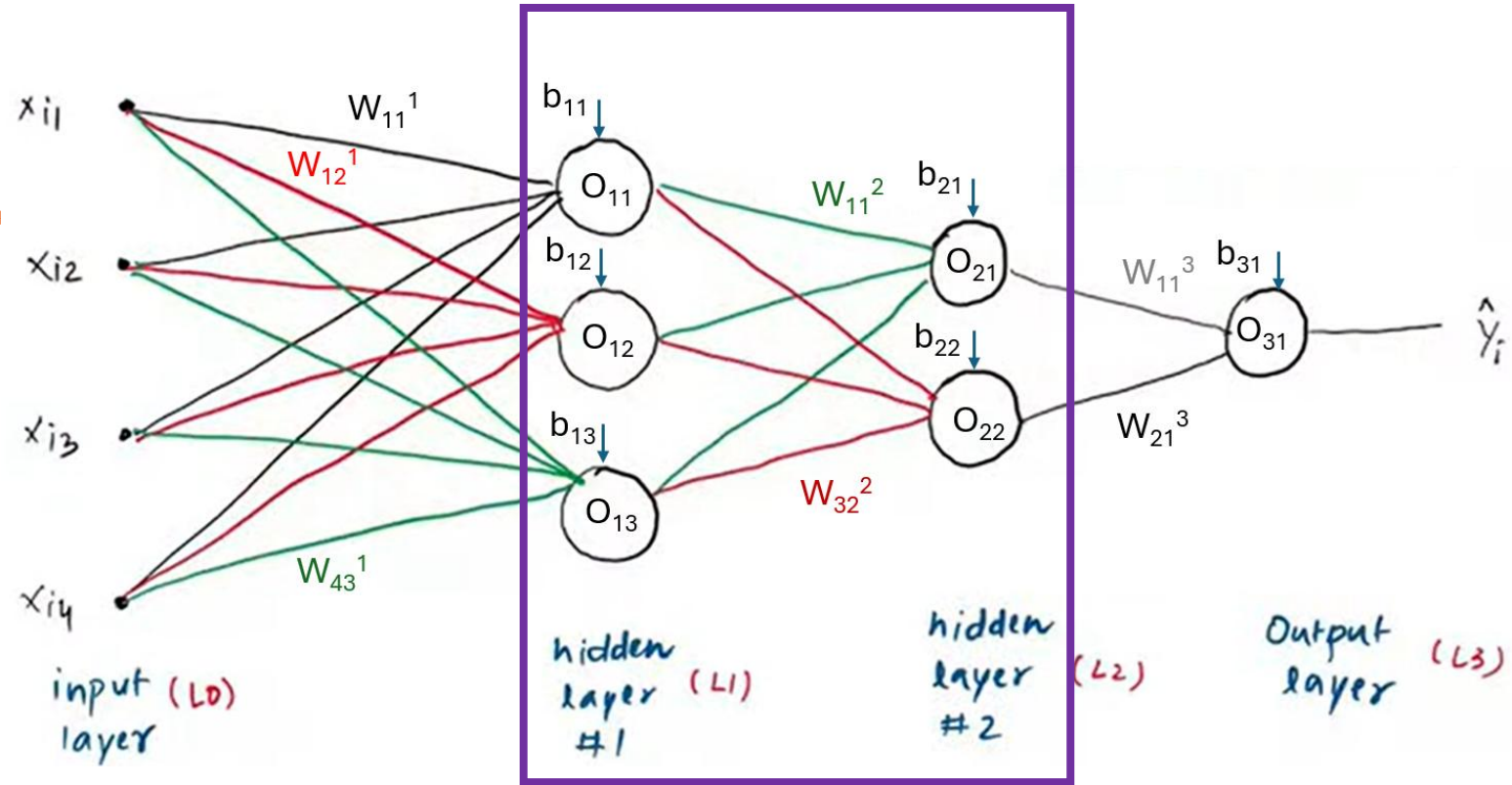
$$\begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} = \sigma \left(\begin{bmatrix} W_{11}^1 X_{i1} + W_{21}^1 X_{i2} + W_{31}^1 X_{i3} + W_{41}^1 X_{i4} + b_{11} \\ W_{12}^1 X_{i1} + W_{22}^1 X_{i2} + W_{32}^1 X_{i3} + W_{42}^1 X_{i4} + b_{12} \\ W_{13}^1 X_{i1} + W_{23}^1 X_{i2} + W_{33}^1 X_{i3} + W_{43}^1 X_{i4} + b_{13} \end{bmatrix} \right)$$

Forward Pass- Layer 2

$$W^{[2]} = \begin{bmatrix} W_{11}^2 & W_{12}^2 \\ W_{21}^2 & W_{22}^2 \\ W_{31}^2 & W_{32}^2 \end{bmatrix} \quad a^{[1]} = \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} \quad b^{[2]} = \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

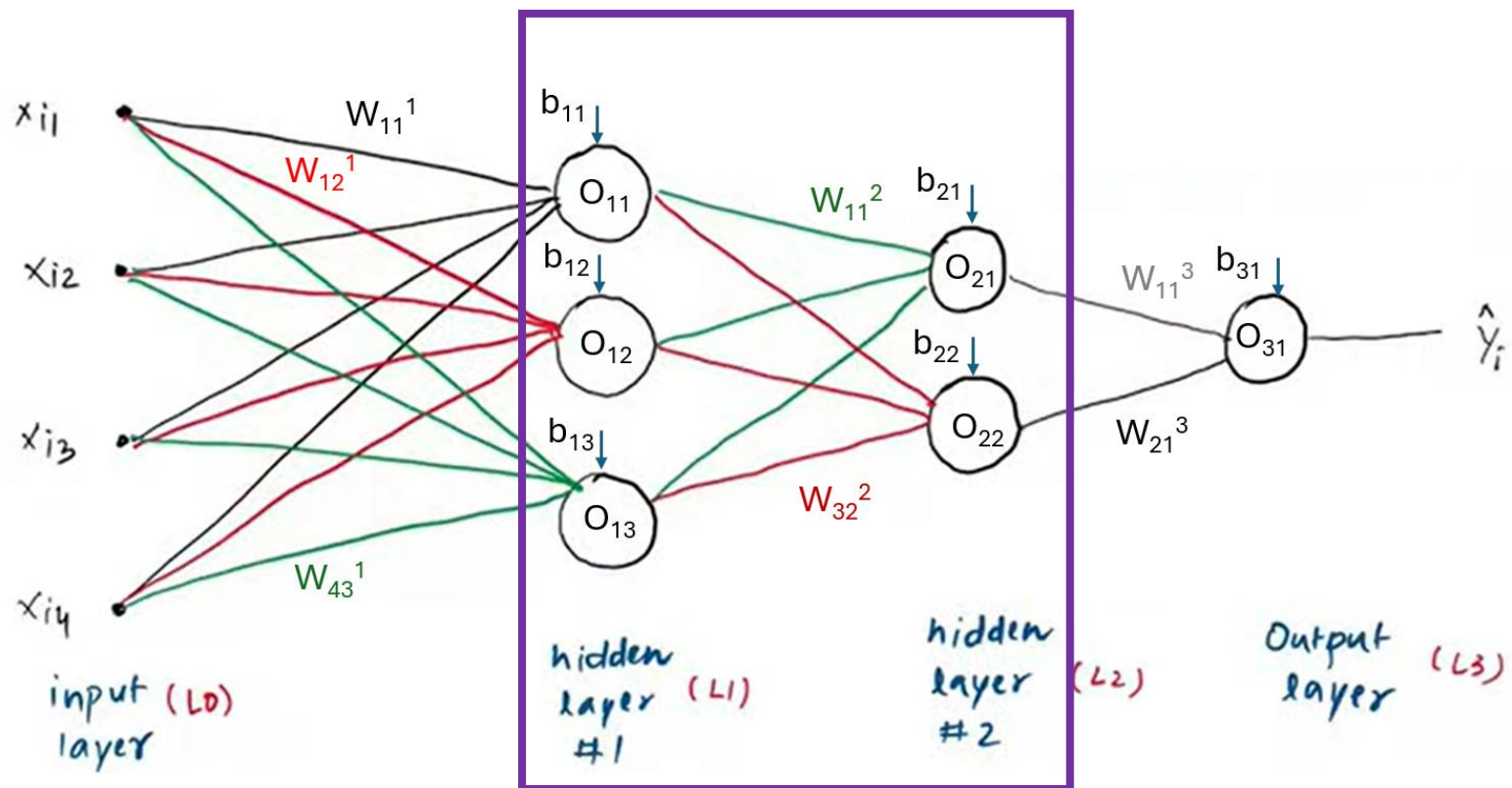
$$\begin{bmatrix} W_{11}^2 & W_{12}^2 \\ W_{21}^2 & W_{22}^2 \\ W_{31}^2 & W_{32}^2 \end{bmatrix}^T \cdot \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$(2 \times 3) \quad (3 \times 1) \quad (2 \times 1)$
 $(2 \times 1) \leftarrow \text{Result}$



$$W^T X + b$$

$$\begin{bmatrix} W_{11}^2 & W_{12}^2 \\ W_{21}^2 & W_{22}^2 \\ W_{31}^2 & W_{32}^2 \end{bmatrix}^T \cdot \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$



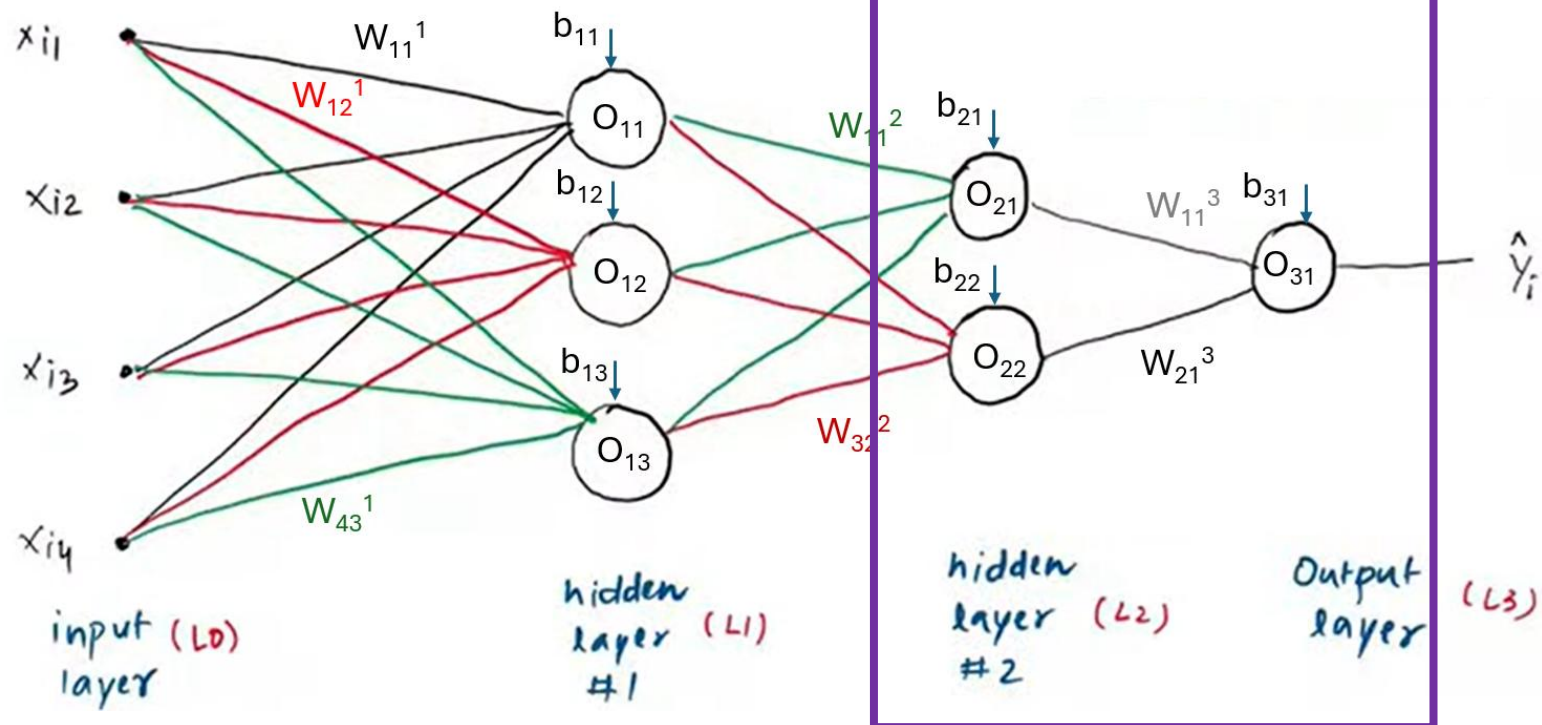
$$\begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} = \sigma \left(\begin{bmatrix} W_{11}^2 O_{11} + W_{21}^2 O_{12} + W_{31}^2 O_{13} + b_{21} \\ W_{12}^2 O_{11} + W_{22}^2 O_{12} + W_{32}^2 O_{13} + b_{22} \end{bmatrix} \right)$$

Forward Pass- Layer 3

$$W^{[3]} = \begin{bmatrix} W_{11}^3 & W_{21}^3 \end{bmatrix} \quad a^{[2]} = \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} \quad b^{[3]} = \begin{bmatrix} b_{31} \end{bmatrix}$$

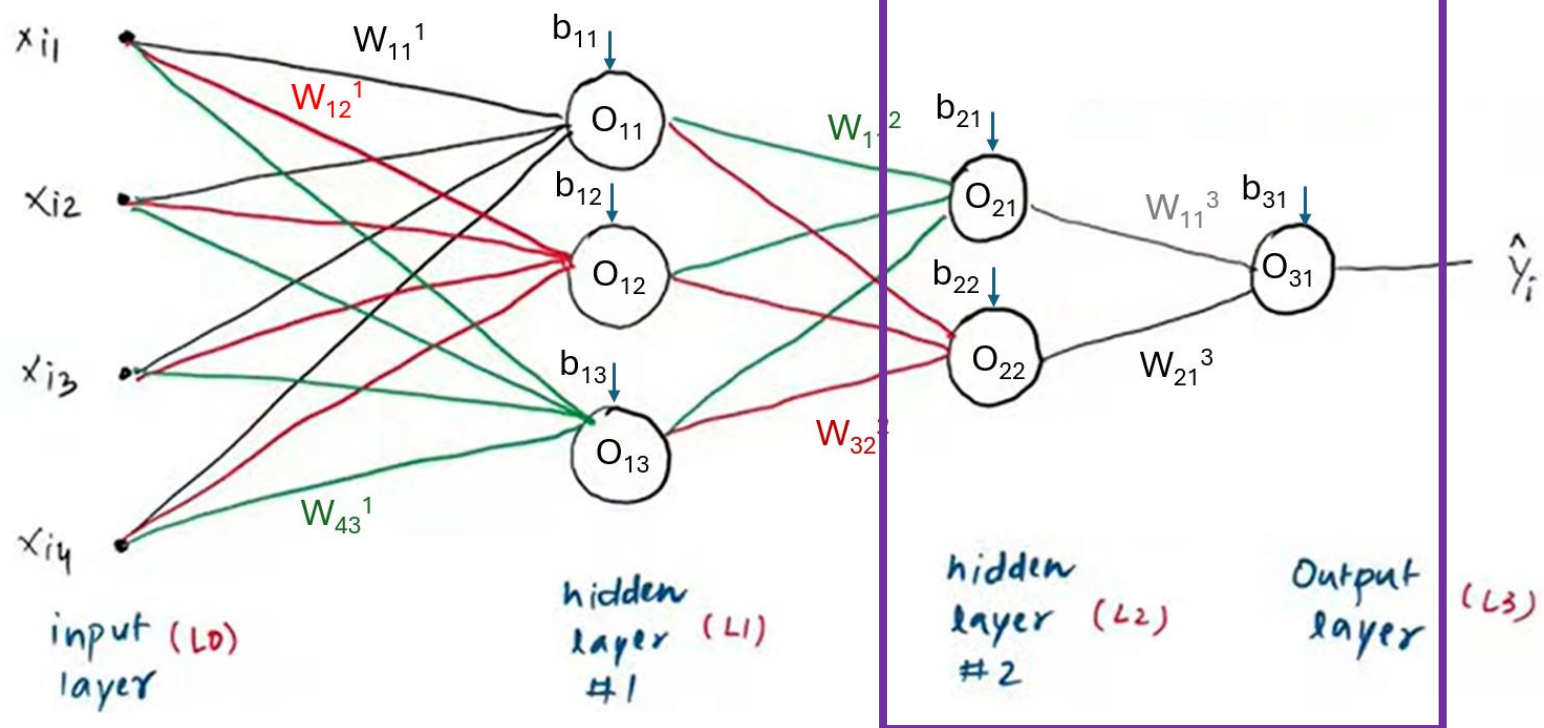
$$\begin{bmatrix} W_{11}^3 & W_{21}^3 \end{bmatrix}^T \cdot \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} + b_{31}$$

$(1 \times 2) \quad (2 \times 1) \quad (1 \times 1)$
 $(1 \times 1) \leftarrow \text{Result}$



$$W^T X + b$$

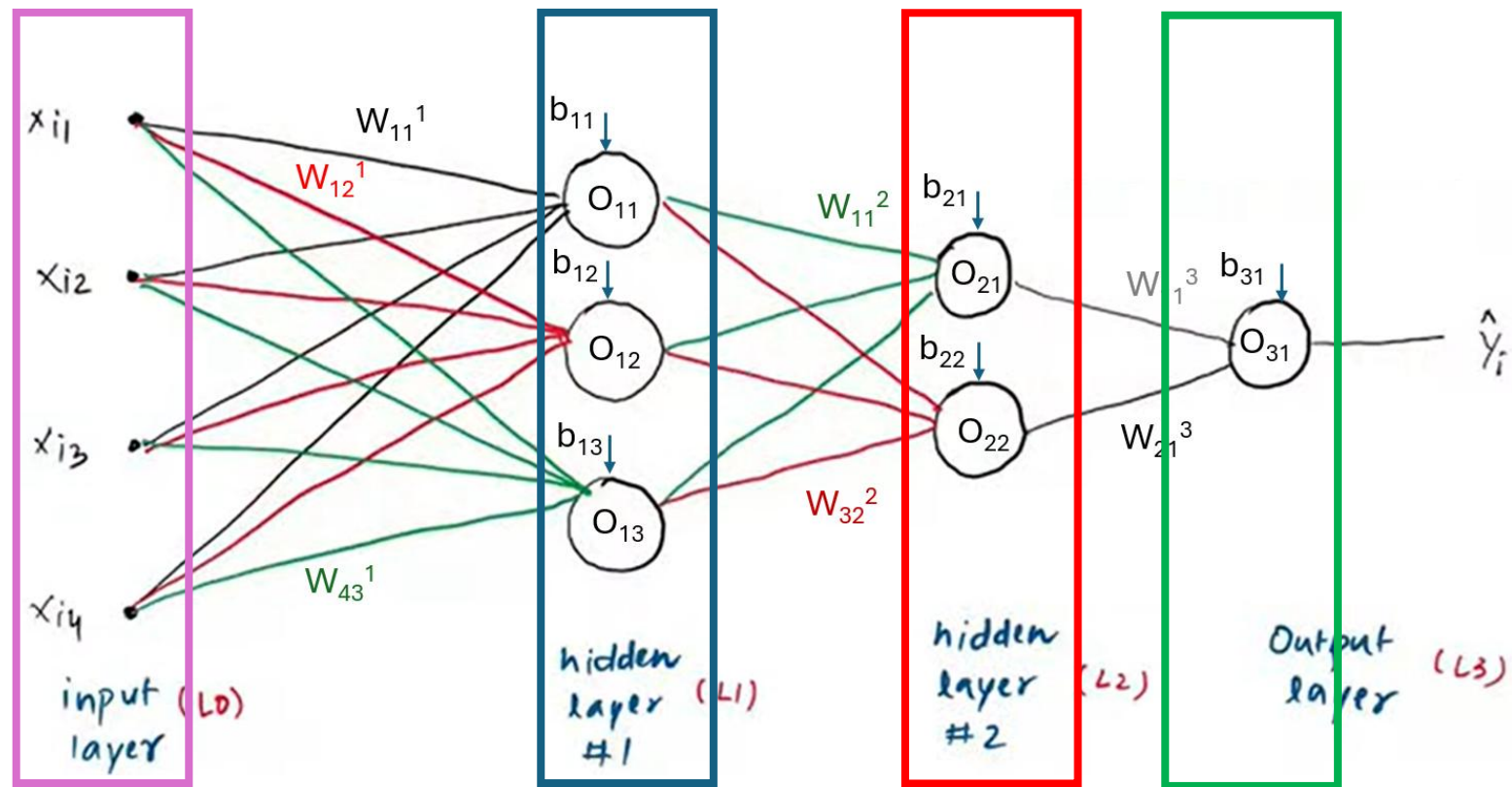
$$\begin{bmatrix} W_{11}^3 \\ W_{21}^3 \end{bmatrix}^T \cdot \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} + b_{31}$$



$$\hat{\mathbf{y}} = \begin{bmatrix} O_{31} \end{bmatrix} = \sigma \left(\begin{bmatrix} W_{11}^3 O_{21} + W_{21}^3 O_{22} + b_{31} \end{bmatrix} \right)$$

Forward Pass

- $a^{[0]} = [X_{i1} \ X_{i2} \ X_{i3} \ X_{i4}]^T$
- $a^{[1]} = \sigma \left((W^1)^T a^{[0]} + b^{[1]} \right)$
- $a^{[2]} = \sigma \left((W^2)^T a^{[1]} + b^{[2]} \right)$
- $a^{[3]} = \sigma \left((W^3)^T a^{[2]} + b^{[3]} \right)$



- $\sigma \left((W^3)^T a^{[2]} + b^{[3]} \right)$
- $\sigma \left((W^3)^T \left\{ \sigma \left((W^2)^T a^{[1]} + b^{[2]} \right) \right\} + b^{[3]} \right)$
- $\sigma \left((W^3)^T \left\{ \sigma \left((W^2)^T \left\{ \sigma \left((W^1)^T a^{[0]} + b^{[1]} \right) \right\} + b^{[2]} \right) \right\} + b^{[3]} \right)$

Quiz # 4-Multi-layer Perceptron

- Apply the forward pass and label the network clearly according to the conventions covered in class. Compute the equations for both outputs.
- Due date: 27-03-2026 (10:00 AM)
- Plagiarism will result in a severe penalty.

